

Knowledge Gap Analysis: Insufficient Evidence for Universal Recycling Feasibility in Allowed Catalog

Input Question

Can we achieve a situation where essentially every material can be recycled and reused?

Domain

Ecology

Validation Status

needs_data

Problem Statement

The question asks whether it is technically and economically feasible to achieve a state where essentially every material can be recycled and reused. Current recycling capabilities are limited by technical constraints (e.g., composite materials, contamination) and economic factors (profitability vs. uninternalized environmental costs).

Rationale

The provided EvidenceCards (EV-Q114-7146a69f82a5c494513af617, EV-Q114-49add93d53207c8ddf6010bf, EV-Q114-86d898aba3878329b030e55d, EV-Q114-548dd4c4bdf946ab806d6965) are exclusively related to telecommunications (MIMO-NOMA, CAP-MIMO, LoRa) and machine learning fairness. They contain no information regarding recycling technology, material science, or environmental economics. Therefore, no grounded scientific hypothesis about recycling feasibility can be generated from the allowed evidence. The research plan must explicitly identify this as a knowledge gap and propose a protocol for external evidence validation rather than asserting unverified facts.

Generated Hypotheses

Hypothesis 1

- **Hypothesis**: [UNGROUNDING_KNOWLEDGE_GAP] Current allowed evidence (MIMO-NOMA, CAP-MIMO, LoRa, ML Fairness) contains zero information regarding recycling feasibility, thermodynamics, or techno-economics; therefore, no system-validatable hypothesis can be generated for Q114 without external evidence ingestion.
- **Mechanism**: The SAGE125 protocol mandates that all factual claims in a hypothesis must be traceable to an allowed EvidenceCard. Since the intersection of {Allowed Evidence Topics} and {Question Domain: Ecology/Recycling} is empty, the mechanism for generating a grounded hypothesis is structurally blocked. Any specific claim about recycling entropy or economics would constitute hallucination relative to the allowed evidence set.

- ****Falsifiable Prediction****: If a valid EvidenceCard pertaining to recycling, LCA, or waste management economics is added to the allowed_evidence_ids list and successfully linked to a factual claim in this hypothesis, then the current 'knowledge_gap' classification is falsified and the hypothesis should be reclassified as 'grounded'.
- ****Required Observations****: Verification that none of EV-Q114-7146a69f82a5c494513af617, EV-Q114-49add93d53207c8ddf6010bf, EV-Q114-86d898aba3878329b030e55d, EV-Q114-548dd4c4bdf946ab806d6965 contain recycling-related content ; Confirmation from EvidenceExtractionResult that established_facts list is empty for ecology domain ; System log confirmation that no new evidence IDs have been authorized since revision_iteration 1
- ****Risk of Being Wrong****: Risk is minimal as this is a meta-statement about system state rather than a scientific claim about the physical world. The only risk is if an allowed EvidenceCard was mislabeled or contains hidden relevant content not captured in the quoted_text field.

Hypothesis 2

- ****Hypothesis****: [EXPLORATORY_EXTERNAL_VALIDATION_REQUIRED] Universal material recycling feasibility cannot be assessed within SAGE125 until evidence coverage is expanded; interim research must be tagged as high-risk exploratory validation requiring external data sources not currently in the allowed catalog.
- ****Mechanism****: Per reviewer feedback (required_revision:d1080301961f), when internal evidence is insufficient, the system may permit gap-driven planning only under explicit 'exploratory' tagging. This hypothesis functions as a procedural placeholder asserting that the research question remains open but unanswerable within current constraints, shifting the verification burden entirely to external validation protocols.
- ****Falsifiable Prediction****: If the system ingests at least one valid EvidenceCard supporting recycling TEA/LCA claims AND the hypothesis is subsequently updated to include that evidence_id in supporting_evidence_ids with a non-zero evidence_support_score, then the 'exploratory_external_validation_required' status is resolved.
- ****Required Observations****: Administrative action adding recycling-relevant EvidenceCards to allowed_evidence_ids ; Successful mapping of new evidence to specific recycling feasibility claims ; Re-evaluation of evidence_grounding_score > 0.0 by ScientificReviewer
- ****Risk of Being Wrong****: High risk that external validation efforts proceed without system oversight, potentially leading to unverified conclusions being treated as system outputs. Also risks indefinite stalling if no new evidence is ever ingested.

Technical Details

This experiment design is strictly classified as [EXPLORATORY_EXTERNAL_VALIDATION_REQUIRED] due to zero evidence grounding in the allowed catalog. The allowed EvidenceCards (EV-Q114-7146a69f82a5c494513af617, EV-Q114-49add93d53207c8ddf6010bf, EV-Q114-86d898aba3878329b030e55d, EV-Q114-548dd4c4bdf946ab806d6965) pertain exclusively to telecommunications (MIMO-NOMA, CAP-MIMO, LoRa) and ML fairness, containing no data on recycling, thermodynamics, or techno-economics. Consequently, this plan does not test a grounded scientific hypothesis but rather establishes a procedural framework for external validation. The technical approach involves defining a meta-protocol for ingesting future LCA/TEA evidence, rather than executing physical or simulation-based experiments on recycling processes. No specific technical claims about entropy costs or NPV are made as facts; they remain unverified knowledge gaps.

Datasets

Source

```
```json
{
 "name": "None (Allowed Evidence Catalog Mismatch)",
 "description": "No valid source datasets exist within the allowed evidence IDs. The current catalog contains only telecommunications and ML fairness papers. Any reference to waste composition databases is currently ungrounded and marked as a knowledge gap.",
 "access_status": "unavailable_in_allowed_evidence",
 "is_downloaded": false,
 "linked_evidence_ids": []
}
```
```

Target

```
```json
{
 "name": "None (Allowed Evidence Catalog Mismatch)",
 "description": "No valid target datasets exist within the allowed evidence IDs. LCA data for composite materials is required but not present in the allowed catalog.",
 "access_status": "unavailable_in_allowed_evidence",
 "is_downloaded": false,
 "linked_evidence_ids": []
}
```
```

Paper Abstract

Background: The question of universal material recycling feasibility requires evidence from material science and environmental economics. Method: We analyzed the allowed EvidenceCards (EV-Q114-7146a69f82a5c494513af617, EV-Q114-49add93d53207c8ddf6010bf, EV-Q114-86d898aba3878329b030e55d, EV-Q114-548dd4c4bdf946ab806d6965) for relevance to ecology. Validation Plan: Establish a protocol for external evidence ingestion. Results: Pending execution of external validation; current internal evidence is insufficient.

Methods

1. Evidence Gap Analysis: Verify that allowed EvidenceCards do not contain recycling-related keywords. 2. Protocol Definition: Define schema for future EvidenceCards required to ground recycling hypotheses. 3. External Validation Tagging: Mark query as requiring external evidence. 4. Risk Assessment: Document hallucination risks if claims are made without evidence.

Experiments

Baselines

```
```json
[
 "Null Baseline: Current system state with zero recycling-related evidence.",
 "Telecommunications Domain Baseline: Verify that existing evidence cards correctly describe MIMO/LoRa systems to confirm domain separation."
]
```
```

Metrics

```
```json
[
 "Evidence Grounding Score (Target: 0.0 for current state)",
 "Domain Mismatch Flag (Binary: True if topic != evidence topic)",
 "Knowledge Gap Count (Number of unverified recycling claims)"
]
```
```

Ablation

Not applicable as no grounded variables exist. The 'ablation' here refers to removing any external knowledge assumptions to ensure strict adherence to the allowed evidence list.

Validation Protocol

Validate that the output explicitly states 'insufficient_evidence' for all recycling claims. Confirm that no factual assertions about recycling feasibility are made without an associated allowed EvidenceCard.

Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。

References

- ****EV-Q114-7146a69f82a5c494513af617**** · arxiv · arXiv:1901.09807
 - authors: Not available · year: Not available
 - url: <https://arxiv.org/pdf/1901.09807.pdf> · doi: Not available
 - reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:1|section:page-1|paragraph:1;
content_sha256=c13a8b582b46941a66e2a0e0ecc12612ab3bb4b14376a7fedd706ba5f5efd91a
- ****EV-Q114-49add93d53207c8ddf6010bf**** · arxiv · arXiv:2301.08411
 - authors: Not available · year: Not available
 - url: <https://arxiv.org/pdf/2301.08411.pdf> · doi: Not available
 - reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:1|section:page-1|paragraph:1;
content_sha256=b6db4a2a9fc6d2ef873d050c54c5e8e3e6f442c0a7a000da13100c641767f00
- ****EV-Q114-86d898aba3878329b030e55d**** · arxiv · arXiv:1904.12300
 - authors: Not available · year: Not available
 - url: <https://arxiv.org/pdf/1904.12300.pdf> · doi: Not available
 - reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:1|section:page-1|paragraph:1;
content_sha256=84c4bf865d77b64f0ee2f51b66f650ff905bd68de3dda48301c8d1c74d336e63
- ****EV-Q114-548dd4c4bdf946ab806d6965**** · arxiv · arXiv:2107.08310
 - authors: Not available · year: Not available
 - url: <https://arxiv.org/pdf/2107.08310.pdf> · doi: Not available
 - reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:1|section:page-1|paragraph:1;
content_sha256=7349e2d63473da75c5f372cd163a24c90a738dd16c4fec922b2e8ad91d7c90d3

Reviewer Comments

- The revision successfully addresses all critical issues from the previous iteration by reclassifying the hypothesis as a '[UNGROUNDING_KNOWLEDGE_GAP]' meta-statement regarding system state, rather than asserting unverified scientific facts about recycling.
- All specific technical claims (entropy costs, NPV, etc.) have been removed from the hypothesis field, strictly adhering to the constraint that factual assertions must be traceable to allowed EvidenceCards.

- The experiment design has been correctly pivoted to an 'Evidence Gap Analysis' and 'Protocol Definition' framework, avoiding any simulation or calculation based on ungrounded external datasets.
- Datasets are now explicitly listed as 'None (Allowed Evidence Catalog Mismatch)' with empty `linked_evidence_ids`, resolving the previous issue of ungrounded dataset references.
- Risk level is appropriately set to 'high' and tagged as 'exploratory_external_validation_required', accurately reflecting the complete lack of internal evidence grounding.
- No Results fabrication detected; status remains pending/not executed. References are null/empty where appropriate, avoiding hallucination.

Revision History

- `auto_revision_1`: 依据评审意见重做假设与实验设计。

Reproducibility Checklist

- Verify that `allowed_evidence_ids` list contains only EV-Q114-7146a69f82a5c494513af617, EV-Q114-49add93d53207c8ddf6010bf, EV-Q114-86d898aba3878329b030e55d, EV-Q114-548dd4c4bdf946ab806d6965.
- Confirm that none of these IDs contain text related to ecology or recycling.
- Ensure that the hypothesis field is tagged as [EXPLORATORY_EXTERNAL_VALIDATION_REQUIRED].
- Check that results field does not contain fabricated numerical values.
- Verify that `execution_metadata.actual_execution` is false.